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Air Quality Index prediction using machine learning for Ahmedabad city

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ABSTRACT

Prediction of air pollution index may help in traffic routing and identifying serious pollutants. Modeling of the complex relationships between these variables by sophisticated methods in machine learning is a promising field. The objective of this work is to compare the various machine learning methods such as SARIMA, SVM and LSTM for the prediction of air quality index for Ahmedabad city of Gujarat, India. In this research, different preprocessing methods are used to manage the data before providing to the machine learning models. This study is carried out based on the data provided by the Central Pollution Control Board of India and it focuses on the support vector machine algorithm with RBF kernel model. So, that the results availed are comparatively better as compared to other kernels of the support vector machine models as well as SARIMA and LSTM models for Ahmedabad city.

1. Introduction

Today air pollution is a major issue in the world. Most of the cities are suffering from air pollution. There are many pollutants like CO, CO2, NO2, NO, SO2, NH3, Particulate Matter (PM 2.5, PM 10), O3, Acrolein, Asbestos, Benzene, CS2, Polycyclic Aromatic Hydrocarbons, Synthetic Vitreous Fibers and total Petroleum Hydrocarbons. Air pollution monitoring has gained attention these days as it has a major impact on the health of humans as well as on the ecological balance. In India, industries are one of the major sources of air pollution, accounting for 51% of total pollution (Express, 2019). The air pollutants index (PM2.5, NO2, SOx, O3 etc.) gives knowledge on how much industries are detrimental to the environment. Air Quality Index (AQI) category taken from Central Pollution Control Board, India (Board, 2019) and its health effects are shown in Table 1. Industrial pollution leads to various respiratory and cardiovascular diseases. The study by Leong et al. (2020) indicates an increase in respiratory disease among workers and their families with an increase in concentration of PM2.5, NO2, and CO. Prolonged exposure of fine matter in particular particles, of less than 2.5 μm in diameter has been shown to increase the relative risk of cardiovascular disease by 1 to 3% (Ejaz et al., 2021). The risk of reproductive system diseases due to air pollution is investigated in the study by Li et al. (2022). The air pollutant (especially NO2) is increasing the risk of reproductive system diseases. In China and India itself WHO estimates that air pollution is responsible for almost seven million annual fatalities (He et al., 2020). There are number of diseases in humans which are directly related with the air pollution such as emphysema, asthma, other respiratory diseases, and chronic obstructive pulmonary disease. The air pollution is not only having an adverse impact on health, but is also affecting the economy severely. According to Zhao (2012), the overall economic impact of particulate matter pollution in 111 Chinese cities in 2004 was US$ 29.18 billion, which was less than the world bank's projection. According to a 2007 world bank assessment (Bank, 2007), the yearly overall health cost associated with air pollution in urban areas of China was estimated to be 157 billion ¥ using human capital approach. According to study (Dong et al., 2021) the GDP per capita growth rate will decrease by 0.05818 percentage points if PM2.5 concentration rises by 1 percent.

2. Related work

The air pollution analysis is considered to be an important study for mapping the pollution level in different cities. The machine learning algorithms are widely used to predict, forecast and control pollution level. Three different machine learning algorithms are discussed as forecasting models for ground level ozone, nitrogen dioxide, and sulfur dioxide by Shaban et al. (2016). The support vector machine (SVM), M5P model tree and artificial neural network (ANN) are used to forecast pollution, the result shows that the best forecasting performance when different features are used in multivariate modeling with the M5P algorithm over other models. Real time sensor data of downtown Toronto, Canada is used by Wang et al. (2020) and they have applied Land Use Regression (LUR), Artificial Neural Network (ANN) and gradient boost machine learning algorithm to forecast. The PM2.5 and Black Carbon (BC) were collected at GPS points using a microaethalometre

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Table 1
India AQI categories and its effect (Central Pollution Control Board, New Delhi, 2020).

API range	Category	Health effects
0–50	Good	Good air quality
51–100	Satisfactory	Low air pollution and no ill effects on health.
101–200	Moderate	Moderate pollution, Breathing discomfort to the people with lungs, asthma and heart diseases
201–300	Poor	Mild aggravation of symptoms among high-risk persons, like those with heart or lung disease.
301–400	Very poor	Significant aggravation of symptoms and decreased exercise tolerance in persons with heart or lung disease.
401–500 or more	Severe	Severe aggravation of symptoms and endangers health

(MicroAeth model AE51). The performance of LUR is observed to be better when the size of the dataset is small, while ANN and gradient boost algorithm gives better accuracy with increase in size of dataset. But the study by Chen et al. (2022) uses 3-dimensional variation approach to integrate multi-source PM2.5 data from satellite and ground sensor data Wuhan, Xiang yang and Yichang (China). It claims better PM2.5 predictions by data assimilation and application of Weather Research and Forecasting model with Chemistry (WRF-Chem) (Grell et al., 2005). Six common pollutants (PM2.5, PM10, SO2, NO2, O3, and CO) and pollution levels are forecasted using an automated air quality forecasting method based on machine learning (Ke et al., 2022). The presented system is used to determine the best model with hyperparameters values without any human interaction. A knowledge base encompassing the meteorological observed data, pollutant concentrations, pollutant emissions, and model reanalysis data was used to assist the integration of five machine learning models and an ensemble model (Stacked Generalization) into the system. The combined air quality benchmark (CLAB) was established for Dallas County (Texas, USA) for average value of benzene, carbon monoxide, nitrogen dioxide, ozone, PM2.5, and sulfur dioxide by Wood (Wood, 2022). Different machine learning (ML) and deep learning (DL) algorithms predict CLAB from 2015 to 2020. Support vector regression (SVR) was observed as best in prediction of CLAB with RMSE of 0.0450 for 2019 data and Adaboost (ADA) with RMSE of 0.0493 for 2020 data of Dallas County. Masmoudi et al. (2020) proposed multi-target regression techniques for the simultaneous prediction of several air pollutant concentrations. It uses a combination of the multi-target regression method and the random forest paradigm. Multi-target Regression exploits correlation between target variables while Random Forest approach is considered better for a traditional single-target prediction task. The novel framework suggested named Ensemble of Regressor Chains-guided Feature Ranking (ERCFR) is applied on UCI dataset and managed to achieve R^2 score of 0.872. The deep learning methods are also employed to predict various pollutants by number of researchers. The Long Short-Term Memory (LSTM) has been used in Li et al. (2017), Xayasouk et al. (2020), Zhao et al. (2019) and Liu et al. (2020) for forecasting the air quality index. LSTM has shown a quick convergence and reduce training cycle with good accuracy (Navares and Aznarte, 2020). Athira et al. use recurrent neural network (RNN), LSTM and Gated Recurrent Unit (GRU) for PM10 forecasting based on the time series from AirNet data (V. et al., 2018). Using different evaluation parameters such as mean squared error (MSE), root mean squared error (RMSE) and mean absolute percentage error (MAPE), authors concluded that the GRU unit performs better than RNN and LSTM unit in forecasting of PM10 on AirNet dataset (Zhao et al., 2018). The Aggregated Long Short-Term Memory (ALSTM) model which is extension of the LSTM model was proposed by Chang et al. (2020). In this proposed model, authors incorporated the stations for regional pollution sources, surrounding industrial regions, and local air quality monitoring stations. The combination of three LSTM models into a predictive model was used for early forecast based on information from adjacent industrial air quality stations and other sources of pollution in order to increase prediction accuracy. Nath et al. compared a deep learning method and statistical method for predicting PM 2.5 and PM 10 (Nath et al., 2021). The statistical model named as Holt–Winters performed better for evaluation measures such as RMSE and MSE based on the available limited data than deep learning methods. In Bekkar et al. (2021) deep learning algorithms like LSTM, Bi-directional LSTM, GRU, Bi-directional GRU, convolutional neural network (CNN), and a hybrid CNN-LSTM models were examined for predicting PM2.5 concentration. The hybrid CNN-LSTM multivariate technique works better in terms of predictive performance and allows for more accurate predictions than any of the classical models on UCI Machine Learning Repository. The Auto Regressive Integrated Moving Average (ARIMA) model is also used to predict air pollution (Gopu et al., 2021). It is a time series analysis model that predicts values based on historical data. The LSTM and ARIMA were compared for forecasting levels of pollution in Mani and Volety (2021). The LSTM was found to be far superior to ARIMA by producing least mean absolute error (MAE) and RMSE values in all the cases on Central Pollution Control Board (CPCB) dataset and data collected from a low-cost IoT hardware setup. Beijing's AQI analysis and AQI forecast in the range from January 2019 to November 2021 is presented in Liu and You (2022). It uses an hourly data of major air pollutants namely PM2.5, PM10, and O3 and average to generate daily data, for further analysis and prediction. The ARIMA and LSTM were used for short-term prediction of Beijing's air quality data.

The overall contribution of this research is described as below.

- The AQI data of Ahmedabad city, Gujarat, India collected from SAFAR (System of Air Quality and Weather Forecasting And Research) (Ministry of Earth Science, Govt. of India.Indian Institute of Tropical Meteorology, 2022) and CPCB (Central Pollution Control Board) (Board, 2019).
- The characteristics of the dataset are analyzed, and data are selected to present effective model learning process.
- Number of data-preprocessing methods are presented to remove the outliers, normalize the data as well as to manage the missing values of the AQI dataset of Ahmedabad city.
- The overall proposed architecture is presented in Fig. 1.
- The various machine learning methods such as SARIMA, SVM with different kernels and LSTM along with hyper parameter tuning are presented to train the machine model for AQI prediction of Ahmedabad city.
- The comparative analysis of these machine learning models is presented, which shows the SVM with radial basis function (RBF) kernel provides better results for AQI prediction using with the help of evaluation measures such as R2 score and RMSE.

3. Data and methods

This section explains the approach used for prediction of AQI. The goal of this study is to use various methods like SARIMA, SVM and

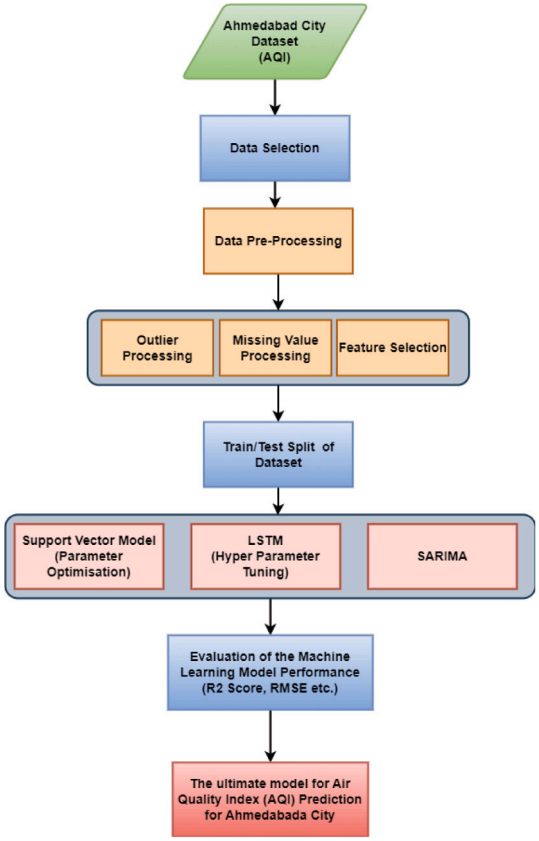

Fig. 1. The overall architecture for AQI Prediction for Ahmedabad City.

LSTM. The procedure includes the following steps: Input Dataset, pre-processing of a dataset, Feature selection,forecasting with ML technique and evaluation of performance. The overview of this study is depicted in Fig. 1.

The detailed description of each stage of the proposed system is described as follows:

3.1. Dataset description

This study is carried out for the city of Ahmedabad (23.0225° N, 72.5714° E), Gujarat from India. Ahmedabad is densely populated (Population: 8,450,228) and the largest industrial center of Western Region of India after Mumbai (Corporation, 2020). The major industries running in Ahmedabad are Textile, Chemical, Manufacturing and Pharma. Lekawada, Raikhad, and Bopal (Industrial areas of Ahmedabad) had some of the highest AQI (AQI 300–400 μg/m3) in India, according to an analysis of SAFAR (System of Air Quality and Weather Forecasting and Research) data (Ministry of Earth Science, Govt. of India.Indian Institute of Tropical Meteorology, 2022). The prediction and planning to control air pollution is required to handle the situation. The data is collected from various stations of Ahmedabad city through central pollution control board and SAFAR presented in Table 2.

The dataset is collected for the duration of January 2015, to January 2021. It is observed from 7 different air quality monitoring stations of Ahmedabad (Board, 2019). The calculation of AQI in India according to updated CPCB air quality criteria uses 12 pollutant indices. CPCB notified updated National Ambient Air Quality Standards (Central Pollution Control Board, New Delhi, 2020) which includes pollutants PM10, PM2.5, NO2, SO2, CO, O3, NH3, Pb, Ni, As, Benzo(a)pyrene, and Benzene. The Actual air pollution concentration (Xi) for every pollutant is normalized in form of sub-index (Ii). Table 3 gives formula of PM10 index (IPM10) for various ranges of PM10 Concentration (ConcPM10.5) in air.

Table 2
List of AQI station of Ahmedabad city.

Sr no.	Station
1	Raikhad, Ahmedabad - IITM
2	Raikhad - SAFAR
3	Navrangpura - SAFAR
4	Phase-4 GIDC, Vatva - GPCB
5	Pirana - SAFAR
6	Maninagar, Ahmedabad - GPCB

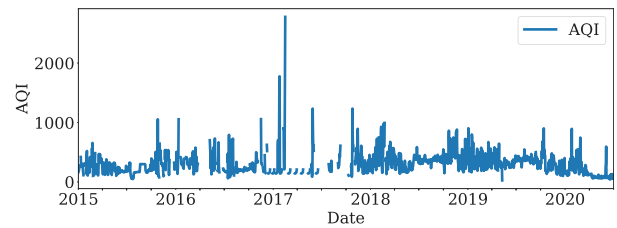

Fig. 2. Time series for Ahmedabad Daily AQI.

The function F combines sub-indices of every individual pollutant for AQI calculation. For the AQI, a maximum operator system has been adopted which is free from ambiguity and eclipsing, as shown below:

$$AQI = Max(I_{PM2.5}, I_{PM10}, \ldots, I_{NH3}) \quad (1)$$

The availability of either I PM2.5 or I PM10 index is mandatory. It is also necessary that at least 3 indices must be available. Fig. 2 presents time series for Ahmedabad Daily AQI data.

3.2. Data preprocessing

Pre-processing contributes in transforming the data into an efficient input format that will be fed to the model. The different pre-processing methods used in this research work are elaborated in this section.

3.2.1. Feature selection

The various pollutant indices CO, NO2, SO2, PM2.5/PM10, O3, NH3 and Pb are used for AQI calculation in India. In the past three years, for the city of Ahmedabad, the exploratory data analysis reveals harmful levels of CO, O3 and PM2.5 and it causes health hazards as stated by Shah and Patel (2021). The dataset provided by CPCB, India contains number of null values and noisy information (i.e. 4681 out of 29531 records). The records for Benzene, Toluene and Xylene (BTX) are not considered for analysis. The availability of CO, O3 and PM2.5 index is better than PM10, SO2 and NO2 index in the dataset. Hence, CO, PM2.5 and O3 index happen to be the key features for AQI prediction of Ahmedabad city.

3.2.2. Outlier processing

The Z-score is one of the most important methods for identifying outliers. A data point's Z-score simply indicates how many standard deviations it is from the mean (Curtis et al., 2016). The threshold value for the Z-score can be established to determine whether to designate a point as an outlier in the overall scheme of things. The outliers can promote modeling error and it could affect the outcomes of the model. The Z-Score represents the standard deviation of the real value from the mean value.

$$z = (x - \mu)/\sigma \quad (2)$$

Where x is the raw score, μ is the population mean, and σ is the population standard deviation.

Fig. 3 presents the AQI data of Ahmedabad city before outlier processing and Fig. 4 presents the AQI data after removal of the outlier using Z-score outlier removal.

Table 3
PM10 index calculation.

PM 10 concentration range (μg/m3)	PM10 index formula
ConcPM10 <= 100	IPM10 = ConcPM10.5
ConcPM10 > 100 and ConcPM10 <= 250	IPM10 = 100 + (ConcPM10 − 100) * 100/150
ConcPM10 > 250 and ConcPM10 <= 350	IPM10 = 200 + (ConcPM10 − 250)
ConcPM10 > 350 and ConcPM10 <= 430	IPM10 = 300 + (ConcPM10 − 350) * (100/80)
ConcPM10 > 430	IPM10 = 400 + (ConcPM10 − 430) * (100/80)

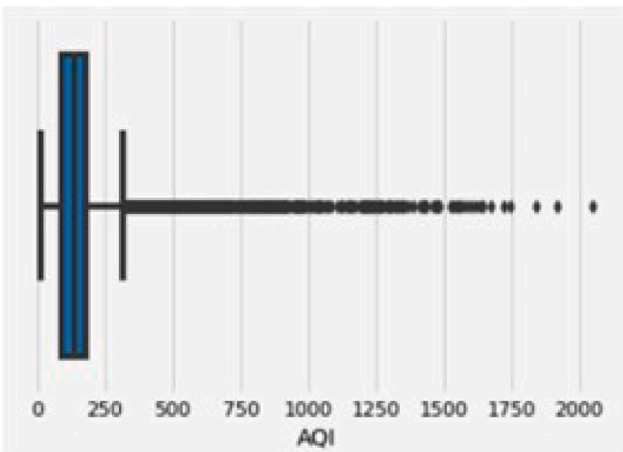

Fig. 3. AQI data before outlier removal.

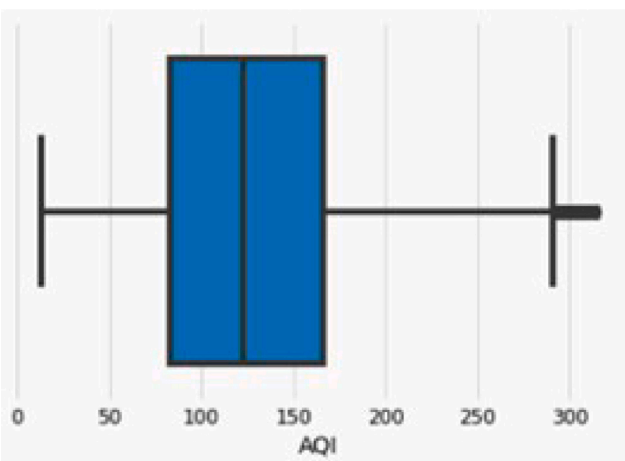

Fig. 4. AQI data after outlier removal.

3.2.3. Missing value handling

There are lots of missing values in present in the dataset of Ahmedabad city. Deleting all rows having invalid or missing values can result in information loss, which would result in imprecise output (Emmanuel et al., 2021). Multiple imputation and maximum likelihood methods (Graham, 2009) are generally used to replace missing values with anticipated values. In this research, mean imputation is used for replacing the missing value. The mean substitution method replaces the variable mean score for missing information. However, a season-specific mean value is used in place of a Null value observed in that season only. The time series of Ahmedabad dataset is shown in Fig. 5 after applying the mean imputation.

3.3. Machine learning models

When considering modeling of air quality, it is challenging to forecast with required accuracy using environment quality model. For Ahmedabad city it is difficult to create a model efficiently for the dynamic behavior of environmental data, particularly when chemical, physical, and/or meteorological data is included for analysis. In order to construct and develop AQI prediction model effectively, three distinct models named as SARIMA, SVM, and LSTM are examined in this research.

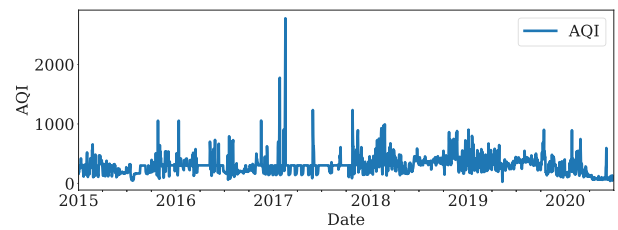

Fig. 5. Time series for Ahmedabad Hourly AQI (After Mean Imputation).

3.3.1. SARIMA

The ARIMA (Auto Regressive Integrated Moving Average) (Chernick et al., 1994; Kumar and Goyal, 2011) refers to a class of models that uses a time series' previous values specifically, its own lags and lagged prediction errors to explain the time series to predict future values. The ARIMA equation has two components AR (Auto Regression) and MA (Moving Average). The predicted value is estimated based on linear combination lags of Y and Lagged forecast errors.

$$Y_t = \alpha + \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + \cdots . + \beta_p Y_{t-p} + \epsilon_t + \phi_1 \epsilon_{t-1} + \phi_2 \epsilon_{t-2} + \cdots . + \phi_q \epsilon_{t-q} \quad (3)$$

where Y_t is observed value of Y at time t, α, β are coefficients and ϵ is error, p is the order (number of time lags) of the AR model, d is the degree of differencing and q is the order of the MA model.

The SARIMA (Seasonal ARIMA) (Dubey et al., 2021) is useful in situations where a time series shows a seasonal variation. The model of SARIMA can be represented by the notations (p, d, q) (P, D, Q)s of an autoregressive notation (P), seasonal moving average notation (Q) and order of difference (D) of seasonal part of the ARIMA model seasonal (Box et al., 2008). The length of the seasonal period is indicated by the subscripted letter "s". The SARIMA can be represented as

$$\phi_p(B)\Phi_P(B^S)(1-B)^d(1-B^S)^D Z_t = \theta_q(B)\Theta_Q(B^S)a_t \quad (4)$$

The ordinary autoregressive regressive component $\phi_p(B)$

$$\phi_p(B) = 1 - \phi_1 B - \phi_2 B^2 - \cdots ....\phi_p B^p \quad (5)$$

The ordinary moving average component $\theta_q(B)$

$$\theta_q(B) = 1 + \theta_1 B + \theta_2 B^2 + \cdots ....\theta_q B^q \quad (6)$$

The seasonal autoregressive component $\Phi_P(B^S)$

$$\Phi_P(B^S) = 1 - \Phi_1 B - \Phi_2 B^{2S} - \cdots ....\Phi_P B^{PS} \quad (7)$$

The seasonal moving average component $\Theta_Q(B^S)$

$$\Theta_Q(B^S) = 1 + \Theta_1 B + \Theta_2 B^{2S} + \cdots ....\Theta_Q B^{QS} \quad (8)$$

where B is the backshift operator, d and D is the non-seasonal and seasonal order of differences. The ordinary autoregressive and moving

average components are represented by polynomials $\phi_p(B)$ and $\theta_q(B)$ of orders p and q. The seasonal autoregressive and moving average components are $\Phi_P(B^S)$ and $\Theta_Q(B^S)$, where P and Q are their orders.

3.3.2. Support Vector Machine (SVM)

One more broadly used machine learning algorithm is SVM (Shalev-Shwartz and Ben-David, 2014). SVM has undergone a remarkable amount of development since Chervonenkis (2013) first introduced it. Support Vector Machines are supervised learning models and the learning algorithms for regression and classification. SVM regression algorithms are commonly referred to as Support Vector Regression (SVR). The straight line needed to fit the data is referred to as the hyperplane in Support Vector Regression. It also works on linearly non separable data by mapping those data into higher order of dimension with the help of different kernel functions named as Radial Basis Function (RBF), Gaussian, Polynomial, etc. SVM has shown state-of-the-art results on different complex situation which are not linearly separable. The number of researchers that have used SVM for the prediction of air pollution (Leong et al., 2020; Ke et al., 2022; Espinosa et al., 2022; Balogun and Tella, 2022).

3.3.3. Long Short-Term Memory (LSTM)

The forecasting model has been developed using Long Short-Term Memory (LSTM) Recurrent Neural Network (RNN) model architecture (Li et al., 2017). The LSTM RNN layers are composed of LSTM units as its building components (Vahora and Chauhan, 2019). The cell state, which retains the data carried by the LSTM unit, is the key component of an LSTM unit. The LSTM unit is controlled by gates, which allow it to add or remove information from its cell state. Additionally, the LSTM unit has a hidden state that keeps track of previous information about the observed sequence. The forget gate, input gate, and output gate are the three gates that make up the LSTM unit. Here, forget gate selects which portion of the cell state should be removed based on the current input, hidden state, and forget gate weights. The input gate calculates the amount of a new input to be added to the cell state using the input gate weights, hidden state, and current input. By using the output gate weights, hidden state, and current input, the output gate determines how the cell state will affect the current output.

3.4. Parameter evaluation and model selection

In this study, R2 Score, Mean Squared Error, and Root Mean Squared Error (RMSE) metrics have been used. These validation metrics are used to determine whether regression models are accurate or misleading.

Mean absolute error (MAE). A measure of errors between paired observations is called mean absolute error (MAE) (Sammut, Claude Webb, 2010). The larger MAE is the larger the error. The error is the difference between predicted value ($\hat{Y}_i$), actual value (Y_i) and n is total number of observation.

$$MAE = \frac{\sum_{i=1}^{n} \left| Y_i - \hat{Y}_i \right|}{n} \tag{9}$$

Root mean square error (RMSE). The measure of how well a regression line fits the data points is called RMSE (Nevitt and Hancock, 2000). RMSE can alternatively be thought of as the residuals' standard deviation. The square root of the mean of the squares of the errors is used to calculate RMSE of projected values for time t of the dependent variable Y_t in a regression using variables observed over T periods.

$$RMSE = \sqrt{\frac{\sum_{t=1}^{T} (\hat{Y}_t - Y_t)^2}{T}} \tag{10}$$

Table 4
ADF test results on Ahmedabad AQI time series.

Test statistic	0.880849
p-value	0.992835
# Lags used	11.00
Number of observations used	55.00
Critical value (1%)	−3.555273
Critical value (5%)	−2.915731
Critical value (10%)	−2.59567

Table 5
White noise test on Ahmedabad AQI time series after differencing.

Difference	X-squared	p-value
6 periods	5.96	9.013×10^{-7}
12 periods	1.47	5.069×10^{-20}

R^2 score. The R^2 Score (Chicco et al., 2021) measures the proportion of variance in situation when independent variables determine the dependent variable. The coefficient of determination (R^2 Score) is more accurate and informative and does not suffer from the interpretability issues. The percentage of the variation in the dependent variable that can be predicted from the independent variable is known as the coefficient of determination, denoted as R^2 or r^2 and pronounced "R squared". The equation for R^2 is given as

$$R^2 = 1 - \frac{\text{sum squared regression (SSR)}}{\text{total sum of squares (SST)}}, R^2 = 1 - \frac{\sum_{i=1}^{m} (Y_i - \hat{Y}_i)^2}{\sum_{i=1}^{m} (Y_i - \bar{Y})^2} \tag{11}$$

Where $\hat{Y}_i$ is the predicted ith value, Y_i element is the actual ith value and $\bar{Y}$ is mean of actual values and m is total number of observations.

3.5. Experimental result analysis and discussion

The experimental results evaluated using three different machine learning algorithms are discussed and comparative analysis is also presented in this section .

3.5.1. SARIMA

The stage wise results of the SARIMA model for AQI prediction are presented in this section.

Statistical test. Seasonality and trends of Ahmedabad daily AQI (January 1, 2015 to January 1, 2021) are plotted in Fig. 6. It presents that earlier in 2015 the AQI value is higher and AQI fluctuated throughout the year. During and after the COVID 19 saturation the AQI value is decreasing. The Augmented Dickey–Fuller (ADF) test has been applied on AQI time series dataset. The results of ADF test on original series are presented in Table 4.

The p-value is less than the significance level of 0.05. It confirms that the series is not a stationary time series. The first order differentiation time series is presented in Fig. 7 for AQI of Ahmedabad city.

Autocorrelation function and partial autocorrelation function plot. There is seasonality in time series. The first-order difference series is analyzed (Annual seasonality lag =12 period). The comparative analysis of ADF tests of first order difference series with 6 period lag and 12 period lag is provided in Table 5.

The results depict that the time series after the first order differentiation (lag=12) is not purely random white noise series and subsequent analysis is required to be performed. The abbreviations ACF and PACF stand for autocorrelation function and partial autocorrelation function, respectively. By analyzing two graphs shown by ACF plot and PACF plot are displayed in Fig. 8 and Fig. 9 respectively, it is determined that which model will be best. Using autocorrelation a time series is calculated and plotted. When the partial autocorrelation represents a correlation between the two variables, the other terms present intervening lags of the model.

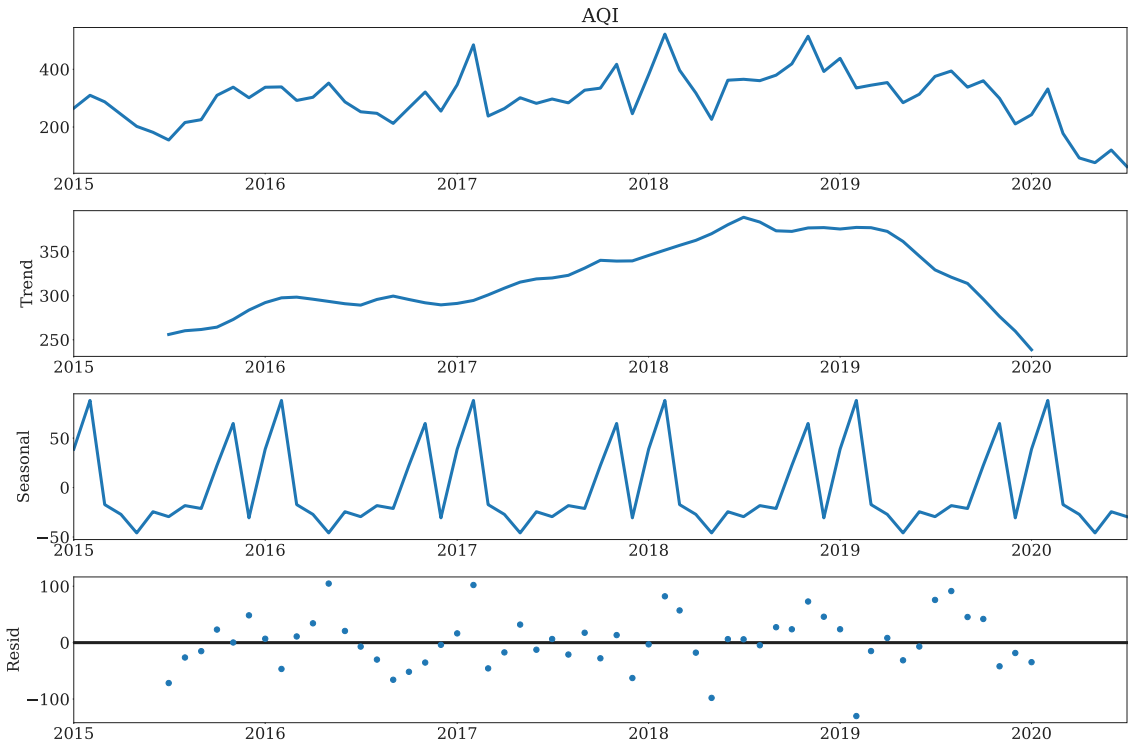

Fig. 6. Seasonality and Trends in Ahmedabad AQI Time Series.

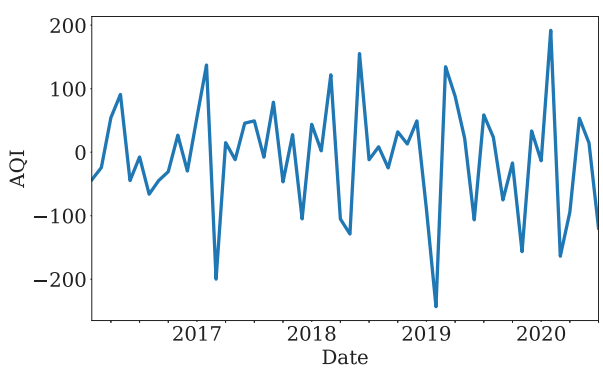

Fig. 7. Time series for Ahmedabad Hourly AQI after First Order Difference.

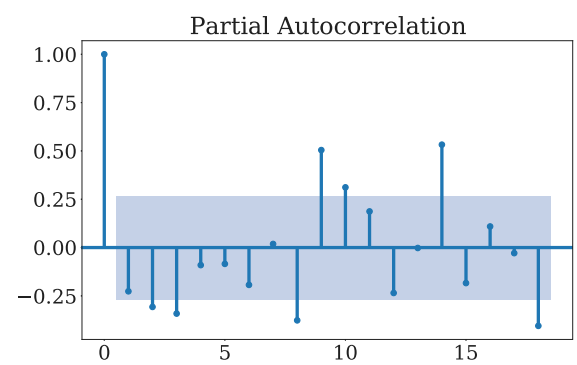

Fig. 9. PACF Plot of Ahmedabad AQI dataset after first order differencing.

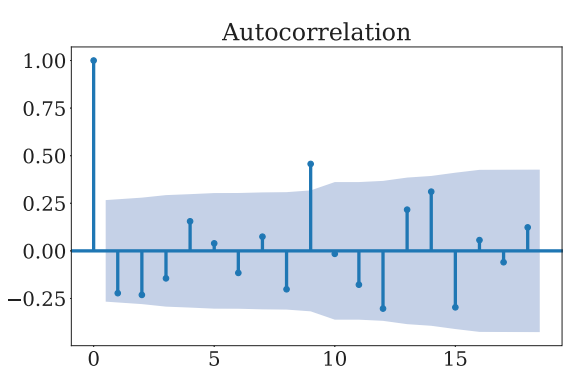

Fig. 8. ACF Plot of Ahmedabad AQI dataset after first order differencing.

Table 6
SARIMA model fitting results.

SARIMA model with parameters	AIC value
SARIMA(0, 1, 0) × (0, 1, 2, 12)12	466.826469
SARIMA(0, 1, 1) × (0, 1, 1, 12)12	440.0162864
SARIMA(0, 1, 2) × (2, 1, 0, 12)12	428.5295059
SARIMA(0, 1, 3) × (2, 1, 0, 12)12	427.9682407
SARIMA(1, 1, 3) × (2, 1, 0, 12)12	430.4331466
SARIMA(2, 1, 2) × (2, 1, 0, 12)12	430.6727208

Test SARIMA with different learning parameters. The identification and order determination of the ARIMA model is intriguing because the autocorrelation coefficient and the order of the partial autocorrelation coefficient of the model after the first-order differentiation are not clear. Therefore, by setting different values of ordinary components of ARIMA(p,q,d) and seasonal components(P,Q,D) are examined. Number of results from all the experimental statistics with lower AIC value (Akaike information criterion) using SARIMA model is presented in Table 6. By using extensive experimental analysis, the results present that model SARIMA (0, 1, 3)x(2, 1, 0, 12)12 provides least AIC Value. AIC evaluates the fitting of the model on the new data rather than present data. The lower AIC value is indicating better fitting of the model (Kletting and Glatting, 2009).

Table 7
Analysis of results different kernel function of SVM for AQI prediction.

Evaluation parameters	Linear	RBF kernel function C = 5000 (regularization constant)	Polynomial C = 1, Degree = 1
Coefficient of determination (R^2)	0.96896094	0.998940471	0.978843972
Mean Squared Error (MSE)	716.100557	24.44433401	488.0897022
Root Mean Squared Error (RMSE)	26.7600553	4.944121157	22.09275225

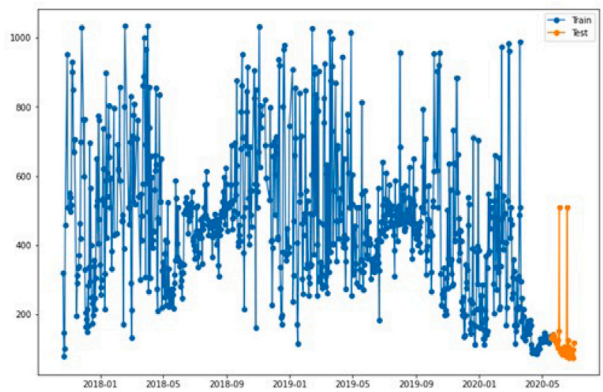

Fig. 10. Splitting data into Train and Test Set.

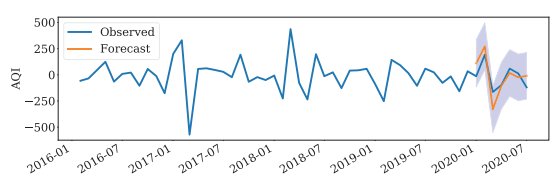

Fig. 11. Comparison of predicted values using SARIMA model versus actual values.

AQI prediction. The Seasonal ARIMA (0, 1, 3)x(2, 1, 0, 12)12 is used to predict AQI from 01-01-2019. The splitting of data into train and test set is shown in Fig. 10. The comparison of predicted values using SARIMA model and actual values is represented in Fig. 11. There are some fluctuations in AQI values due to low accuracy in prediction. However, the result of the proposed model is better in terms of the evaluation parameters MAE =18.67 and RMSE =29.71 compared to the results presented in Liu and You (2022).

3.5.2. SVM

The support vector regression model transforms linearly separable data into high dimensional space using kernel functions (Dun et al., 2020). In order to analyze the behavior of the data and performance of the different SVM kernel function on these data, the predicted value of each kernel function is analyzed. The outcome of the three different kernel functions of the SVM model linear, RBF and polynomial is presented in Table 7.

3.5.3. LSTM

Moreover, the LSTM RNN model has shown state-of-the-art results in AQI prediction (Seng et al., 2021), due to long range dependency of the dataset. The precise model is adopted, which was developed and trained having two layers as 1. LSTM layer with 256 cell along with Relu activation function and 2. Dense layer with one neuron along with linear activation function. The other parameters of the proposed model are batch size is 64, learning rate is 0.001 and Adam optimizer. Fig. 12 presents the train loss and test loss over 500 epochs using on this configuration. The overall results of the LSTM model are presented in Table 8.

Figs. 13 and 14 offer a comparative analysis of the various models, including SARIMA, SVM with different kernel functions, and LSTM.

Table 8
Result analysis of LSTM model for AQI prediction.

Evaluation parameters	LSTM model
Coefficient of determination (R^2)	0.951236947
Mean Squared Error (MSE)	656.6232392
Root Mean Squared Error (RMSE)	25.62466076

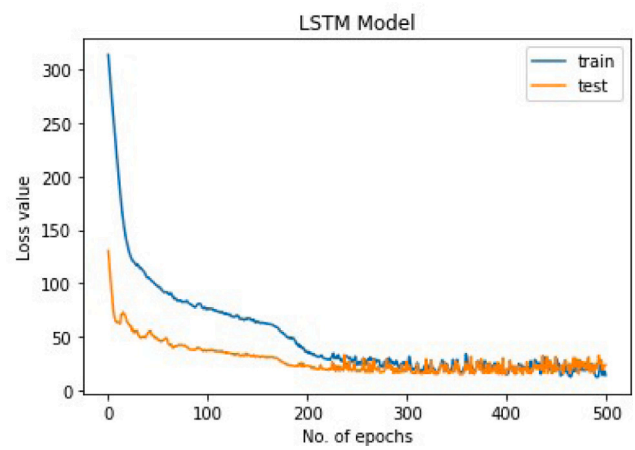

Fig. 12. Loss function curve on AQI prediction of Ahmedabad city.

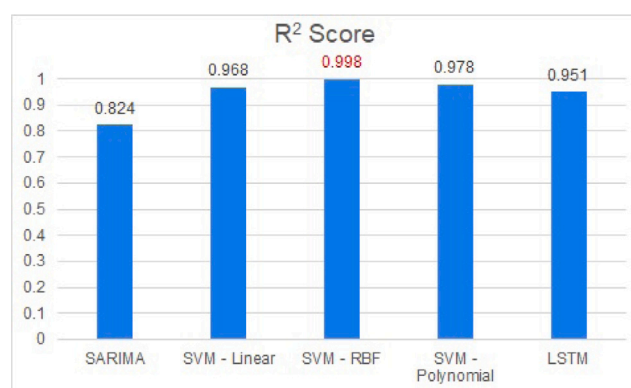

Fig. 13. Comparative analysis of R2 score of different models.

As compared to other SVM kernels, LSTM, and SARIMA models, SVM with RBF keener provides superior results (Vogel and Meghana, 2021; Anyanwu et al., 2023).

4. Conclusion

In this study, an efficient model for the air quality index prediction is presented using the air quality dataset for Ahmedabad provided by SAFAR and CPCB. The different pre-processing methods are used for improved data representation such as outlier removal, data normalization, feature selection and missing value management. Three different machine learning models—SARIMA, SVM with various kernel functions, and LSTM with various hyper parameter tuning—have been investigated in this study to produce an effective predictive model.

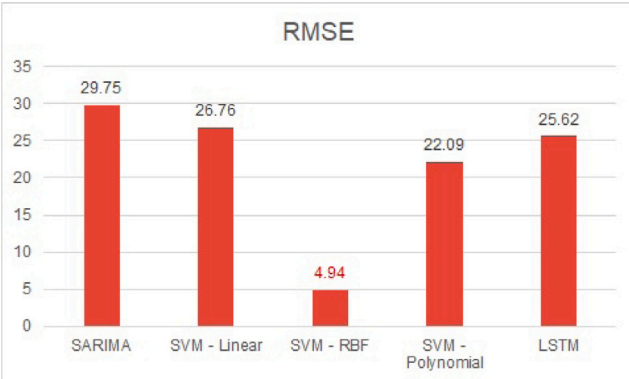

Fig. 14. Comparative analysis of RMSE score of different models.

When predicting the AQI data for Ahmedabad city, the SVM with RBF kernel outperformed other models and various SVM model modifications on the multiple assessment metrics known as R2 score and RMSE. The proposed model can be investigated in the future for the prediction of AQI data in various geographic regions, such as city- or state-level analysis. It can also be expanded to forecast other pollution indices at different levels.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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